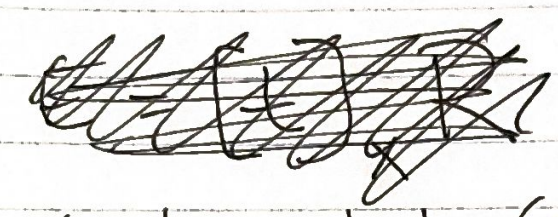
Assumption: camera 1 (Left) is at the origin w/ no rotation or translation

① Epipolar Constraint (From Module 2)



Essential matrix (E) for the 2-camera system can be found from the translation and rotation between the two cameras, and vice versa, using SVD (Singular Value Decomposition).

$$E = [t]_x R$$

The 3D coordinates of a point in camera 1 view and that same point with different 3D coordinates in camera 2 view, along with the Essential matrix, multiplied together, equal zero.

$$X_L^T E X_R = 0$$

② we do not know X_L or X_R. So we convert these to image coordinates using perspective projection equations.

$$u = f_x \frac{X}{Z} + O_x \quad , \quad v = f_y \frac{Y}{Z} + O_y$$

To Matrix Form.

$$X_L^T = [u_L, v_L, 1]\, Z_L\, K_L^{-T}$$

$$X_R = K_R^{-1} Z_R [u_R, v_R, 1]$$

$$[u_L, v_L, 1]\, Z_L\, K_L^{-T} E\, K_R^{-1} Z_R [u_R, v_R, 1] = 0$$

$Z_L, Z_R \neq 0$, we can divide them out

$$[u_L, v_L, 1]\, K_L^{-T} E\, K_R^{-1} [u_R, v_R, 1] = 0$$

$$F = K_L^{-T} E\, K_R^{-1} \quad \text{(Fundamental Matrix)}$$

$$[u_L, v_L, 1]\, F\, [u_R, v_R, 1] = 0$$

$$E = K_L^T F K_R \quad \text{(Essential Matrix)}$$

③ Estimate Fundamental Matrix

8 (at least)

a. find corresponding image coordinates from both camera views, for 1 frame.

b.
$$\begin{bmatrix} u_{L,1}, & v_{L,1}, & 1 \\ \vdots & & \\ u_{L,8}, & v_{L,8}, & 1 \end{bmatrix} F$$

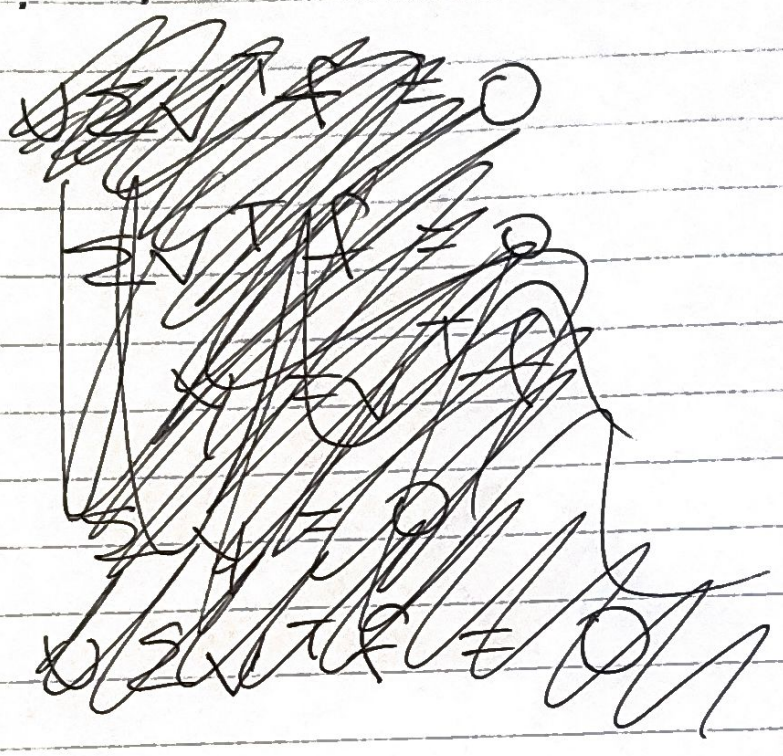
$$\begin{bmatrix} u_{R,1}, & v_{R,1}, & 1 \\ \vdots & & \\ u_{R,8}, & v_{R,8}, & 1 \end{bmatrix} = 0$$

c. expand and rewrite and use SVD.

$$\underset{8\times 9}{A}\ \underset{9\times 1}{f} = 0, \quad \|f\| = 1$$

constrained linear least squares

d. $A = U \Sigma V^T$



f = last column of V

$f = [f_1 \dots f_9]$

$$F = \begin{bmatrix} f_1 & f_2 & f_3 \\ f_4 & f_5 & f_6 \\ f_7 & f_8 & f_9 \end{bmatrix} \quad 3 \times 3$$

Note: My research has led me to understand a second SVD is normally applied to F before using

recompose into F_2 matrix

* $F_2 = U_F \Sigma_F V_F^T$
* smallest value of Σ_F set to 0.
* $F_2 = U_F \Sigma_F' V^T$

(4) [illegible]

$P_L = [I \mid 0]$ } Camera 1 (Left) at origin

$P_R = [[e_2]_x F \mid e_2]$

e_2 = projected center of Left camera, seen in Right image

$F^T e_2 = 0$ } use SVD

$F = U \Sigma V^T$

e_2 = last column of U

$$[e_2]_x = \begin{bmatrix} 0 & -e_z & e_y \\ e_z & 0 & -e_x \\ -e_y & e_x & 0 \end{bmatrix}$$

(5) Crossproduct of u_i and point X_p is zero

$$u_L \times (u_L X_p) = 0$$

$$u_R \times (u_R X_p) = 0$$

expand and reform

$$A X_p = 0$$

use SVD...

$$A = U \Sigma V^T$$

X_p = last column of V

⑥ $X_E \approx H X_p$

expand + form new matrix A

$Ah = 0$

use SVD, $A = U \Sigma V^T$

h = last column of V

$H =$ $\begin{bmatrix} h_1 & h_2 & h_3 & h_4 \\ h_5 & h_6 & h_7 & h_8 \\ h_9 & h_{10} & h_{11} & h_{12} \\ h_{13} & h_{14} & h_{15} & h_{16} \end{bmatrix}$

$X_{E,target} = H X_p$

$= [X_w, Y_w, Z_w, W_w]^T$

(7) Real world Coordinates

$$X_{world} = \frac{X_w}{W_w}$$

$$Y_{world} = \frac{Y_w}{W_w}$$

$$Z_{world} = \frac{Z_w}{W_w}$$

(8) using euclidean distance between two points, we can find dimensions

$$d = \sqrt{(X_B - X_A)^2 + (Y_B - Y_A)^2 + (Z_B - Z_A)^2}$$